Craftax runs compared: achievements and what unlocks them

The map-model (RSSM-2) family and the plain baselines: 1.1M env steps each, size50m, scored on the same 30 fixed worlds with tools/death_eval.py. Rows with several training seeds show seed 0. Updated 2026-11-08.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). Every checkpoint is scored on the same fixed worlds (env.reseed, audit B05), so comparisons are paired world by world. The error bars cover the worlds only, not training seeds, and one 30-world score moves by about ±0.55 between world sets: a gap under ~1.1 between two single runs is not evidence.

No run here uses the rule mask or any other game rule fed to the agent. The map targets (map runs) and the tech-tree potential (potential runs) are hand-built from game knowledge at training time only; the memory target (abl_rssm2) uses none.

Best scores: the honest map (observation-only targets, potential shaping) 6.17 ± 0.34 on the old code and 5.70 ± 0.21 rerun on the fixed code; the privileged-map control also 6.17 ± 0.23. On the fixed code the honest map leads vanilla DreamerV3 by +1.07 ± 0.24 (paired), though that run also has the potential shaping. RSSM-2 as a memory alone (abl_rssm2) scores 5.33 ± 0.22, +0.57 ± 0.31 over plain (+0.57 ± 0.26 against plain's two seeds): not established, and it trains slightly below plain (5.18 vs 5.43 over the last 100k steps).

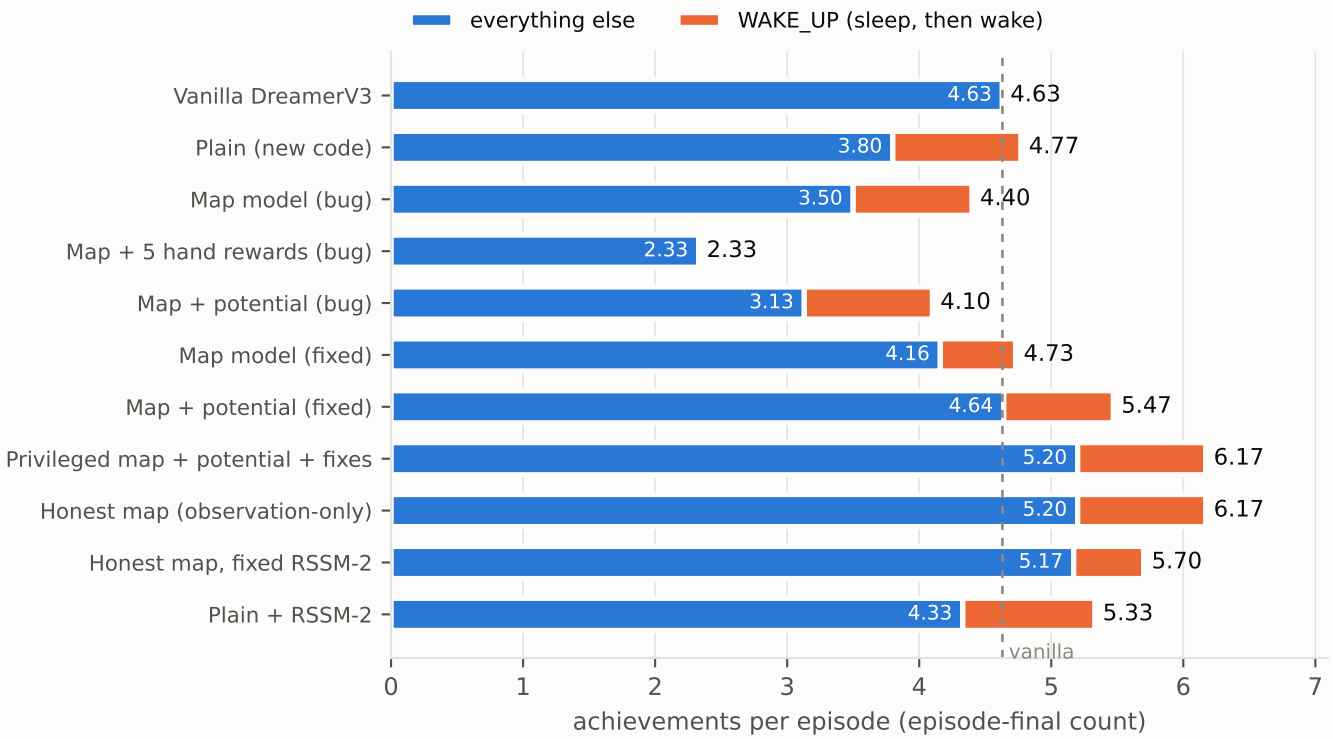
WAKE_UP (sleeping through the night) is a large share of every recent score, plain included (97% of plain's episodes, none of old vanilla's). Without it: vanilla 4.63, plain 3.80, map alone 4.16, honest map 5.20, its fixed rerun 5.17, abl_rssm2 4.33. Among the map runs the potential raises the wood pickaxe from 10% of episodes (map alone) to 33-40% (map + potential, honest map); vanilla made it in 23%, plain in 3%. Normalized return: 2.33% for the honest map, 2.12% for its fixed rerun, 1.96% for abl_rssm2, 1.71% for plain and 1.65% for vanilla.

run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Plain (new code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%
Map model (bug)	4.40 ± 0.29	3.50	263	1.55%	40%	0%	0%	0%	47%
Map + 5 hand rewards (bug)	2.33 ± 0.19	2.33	265	0.63%	0%	0%	0%	0%	43%
Map + potential (bug)	4.10 ± 0.19	3.13	275	1.42%	17%	0%	0%	0%	43%
Map model (fixed)	4.73 ± 0.29	4.16	276	1.70%	67%	10%	0%	0%	77%
Map + potential (fixed)	5.47 ± 0.34	4.64	264	2.02%	70%	33%	13%	17%	60%
Privileged map + potential + fixes	6.17 ± 0.23	5.20	304	2.33%	97%	0%	50%	0%	80%
Honest map (observation-only)	6.17 ± 0.34	5.20	280	2.33%	73%	40%	17%	7%	77%
Honest map, fixed RSSM-2	5.70 ± 0.21	5.17	268	2.12%	87%	7%	50%	0%	77%
Plain + RSSM-2	5.33 ± 0.22	4.33	251	1.96%	80%	0%	27%	0%	77%

ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Vanilla: the original September checkpoint. Plain (new code): main's code. Map rows before honest map fix ran on code older than the 2026-10-11 RSSM-2 fixes (docs/rssm2-step-alignment.md); they are the original checkpoints, re-scored. 'bug' = imag_shift True (the crop slides through imagination; broken, fixed 2026-09-20). map: RSSM-2 trained on a coarse map of the level, read by the actor and critic. 5 hand rewards: survival 'shaped'. potential: survival 'potential' (tech-tree spine). Privileged map: targets from the true map on every cell (the old runs' targets), with the recipe and death fixes to the potential. Honest map: targets only from what the agent observed, with hindsight. Plain + RSSM-2: RSSM-2 as a memory (no map labels), no shaping.

Achievements per run, and the early tech tree

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs

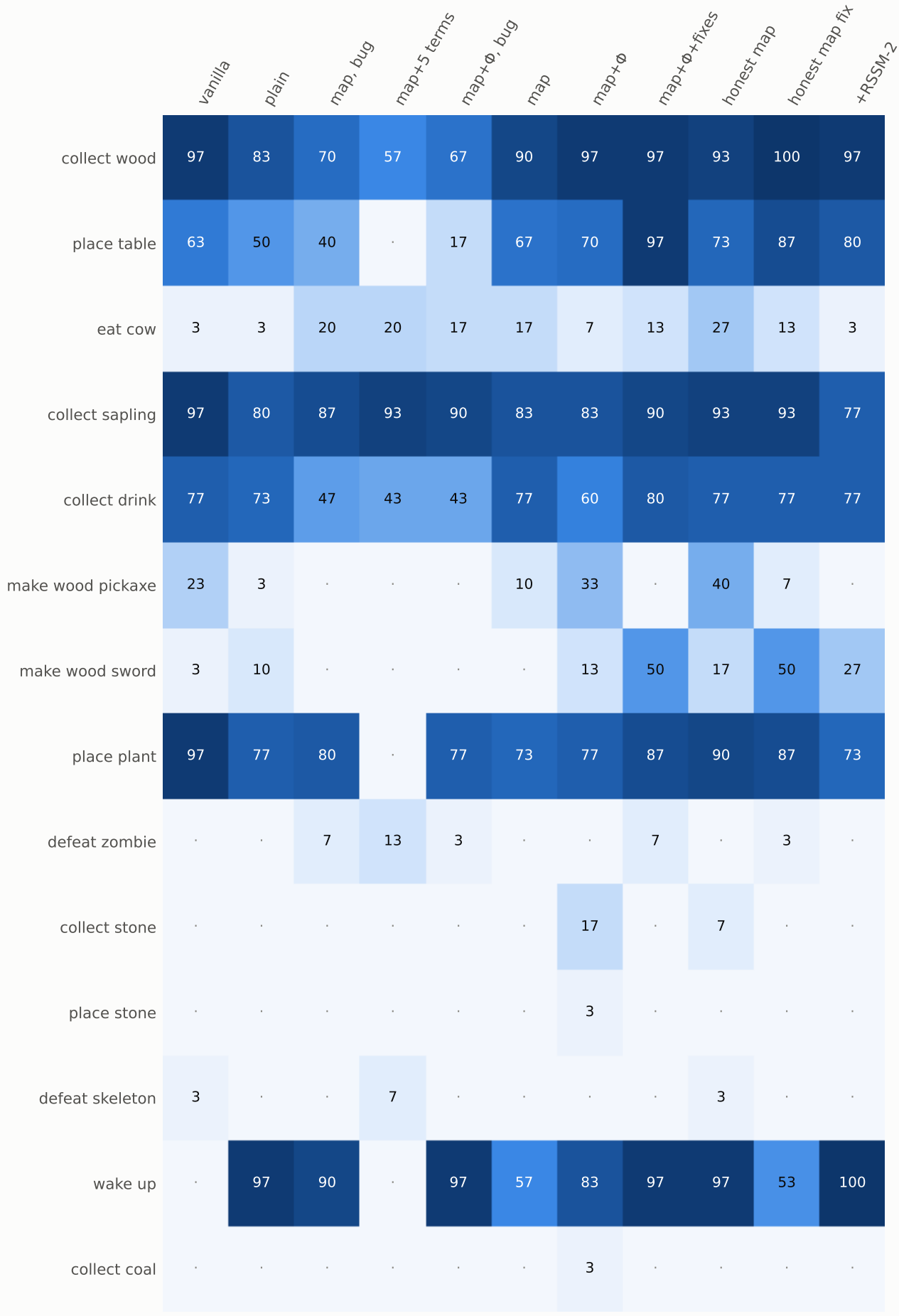


Unlock rate (% of episodes), early tech tree in dependency order

	collect wood	place table	make wood pickaxe	make wood sword	collect stone	place stone	place furnace	collect coal	make stone pickaxe
Vanilla DreamerV3	97	63	23	3	0	0	0	0	0
Plain (new code)	83	50	3	10	0	0	0	0	0
Map model (bug)	70	40	0	0	0	0	0	0	0
Map + 5 hand rewards (bug)	57	0	0	0	0	0	0	0	0
Map + potential (bug)	67	17	0	0	0	0	0	0	0
Map model (fixed)	90	67	10	0	0	0	0	0	0
Map + potential (fixed)	97	70	33	13	17	3	0	3	0
Privileged map + potential + fixes	97	97	0	50	0	0	0	0	0
Honest map (observation-only)	93	73	40	17	7	0	0	0	0
Honest map, fixed RSSM-2	100	87	7	50	0	0	0	0	0
Plain + RSSM-2	97	80	0	27	0	0	0	0	0

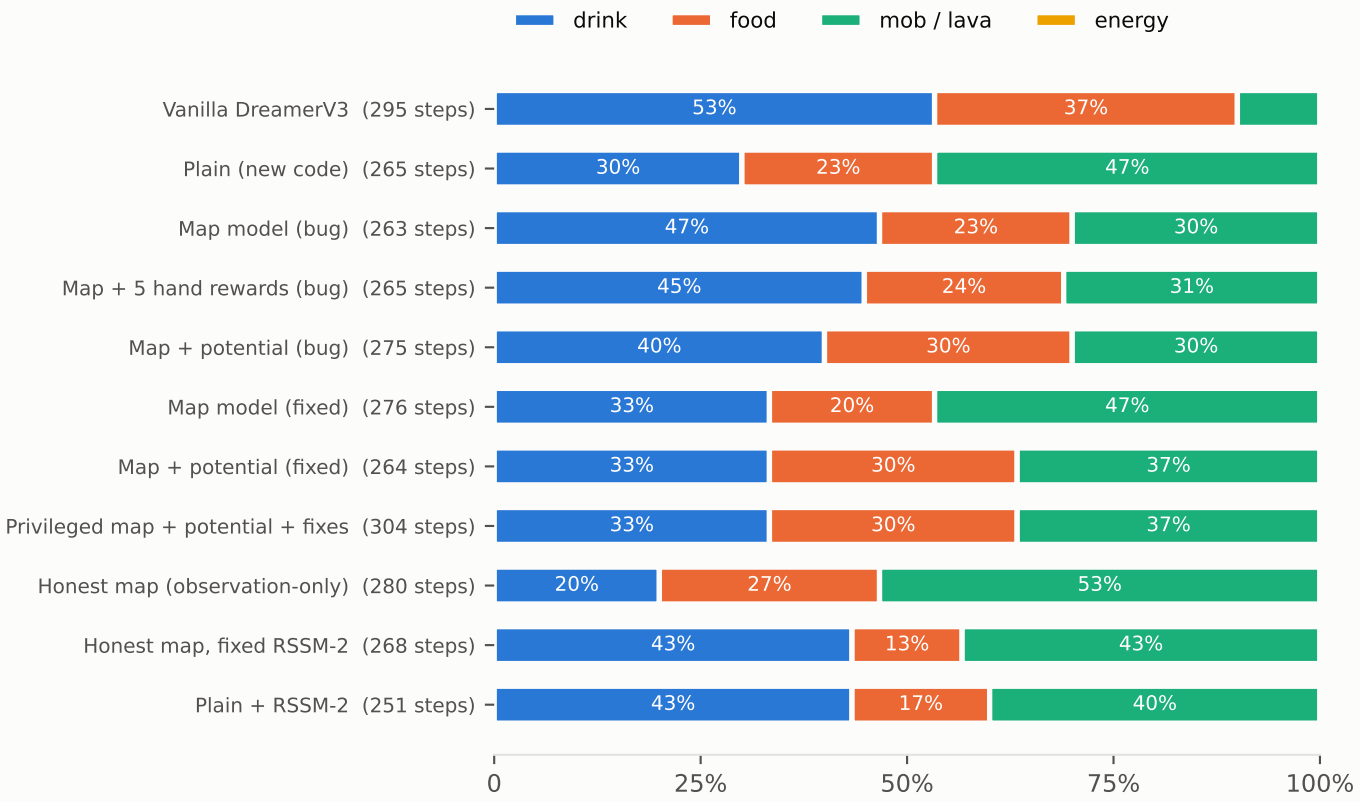
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 53 of 67 were never unlocked by any run.



Cause of death and lifespan

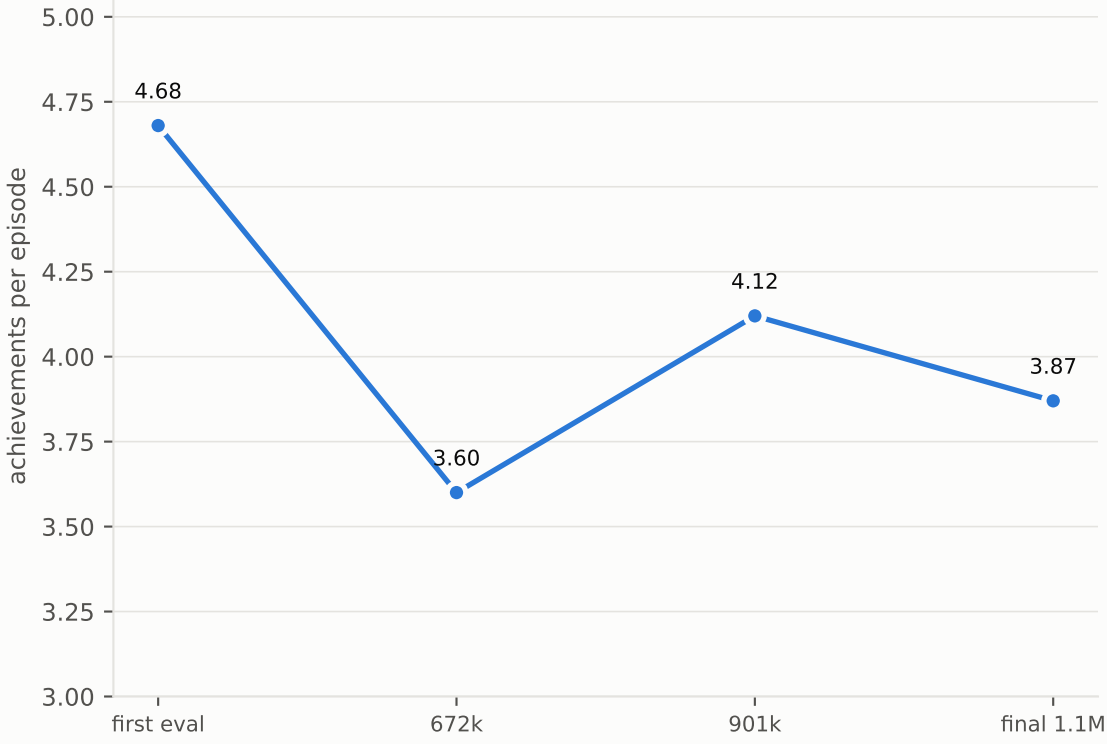
Cause of death (share of evaluation episodes) and lifespan; and, if the results list one, how much a run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333.

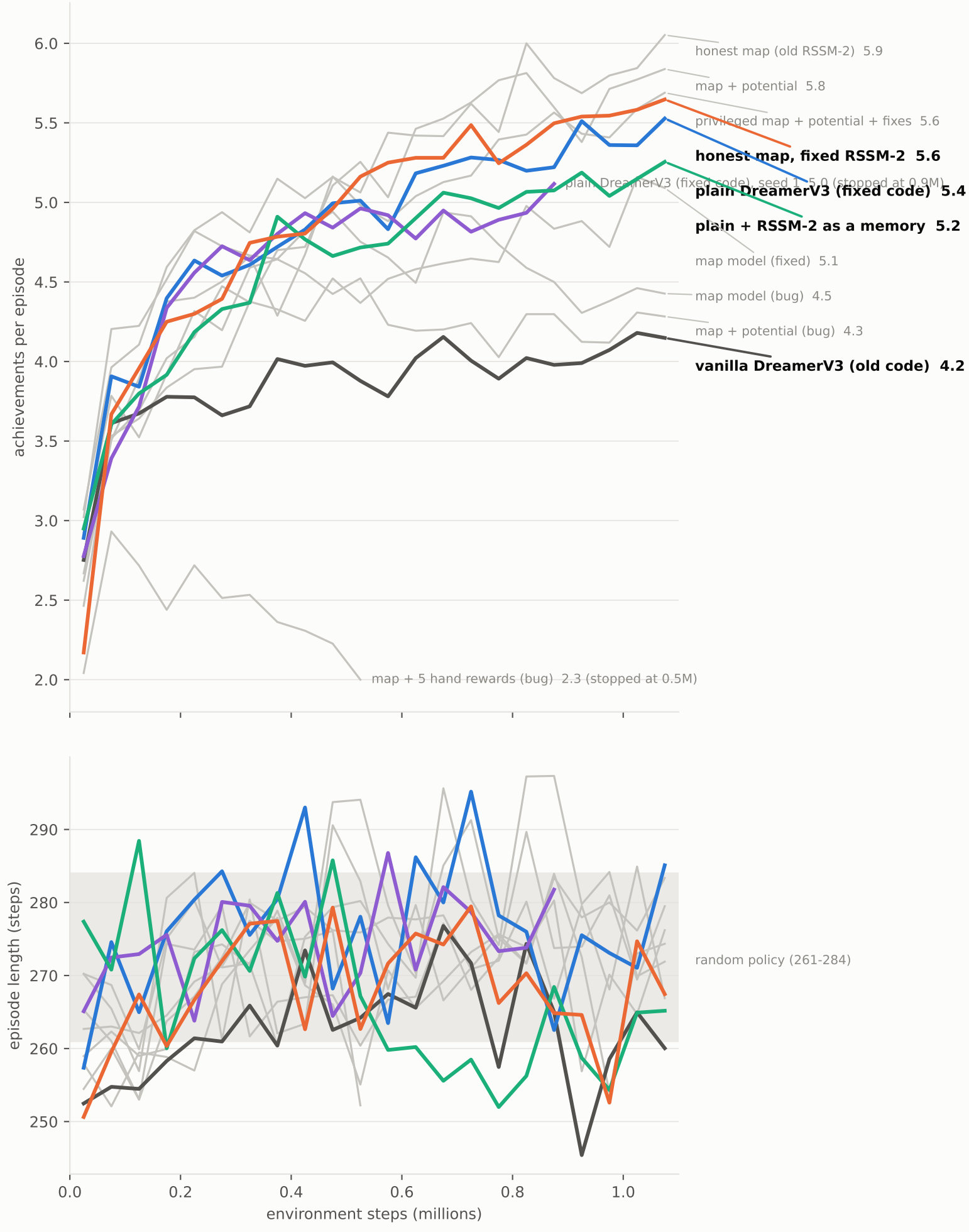
One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each, on the evaluation protocol before the fixed worlds), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps.



What the results say, and what would settle the rest

1. A map helps a little; the potential shaping helps more

With the imagination bug fixed, the map model alone (map, fixed: 4.73) is level with vanilla (4.63) and plain (4.77). Adding the tech-tree potential lifts it to 5.47, and the recipe and death fixes to the potential to 6.17 (privileged control). Among the map runs the potential is what makes the wood pickaxe: 33% of episodes with it against 10% without (the privileged control, oddly, never makes one while placing a table in 97%). Without WAKE_UP the honest map still leads plain by 1.4 achievements.

2. Observation-only targets cost nothing

Training RSSM-2 only on what the agent has seen (honest map, 6.17) matches the privileged control trained on the true map of every cell (6.17), and the honest run makes the wood pickaxe in 40% of episodes where the control never does. So the full-map labels were not what the map model needed.

3. The RSSM-2 fixes did not explain the result

The 2026-10-11 audit found three RSSM-2 bugs, including training seeing up to 7 future steps. The honest map rerun with every fix scores 5.70 against 6.17: within the noise of one score. The map model's contribution was not driven by the leak.

4. RSSM-2 as a memory alone is not established

On plain DreamerV3 with no shaping, RSSM-2 as a memory (predict RSSM-1's latent ahead, recall the observation back) adds +0.57 on the test worlds but trains slightly below plain. One seed; the two measures disagree, which is what noise looks like. It is the memory mode the goal-manager families build on.

5. Hand-written survival rewards hurt

The five survival terms ('shaped') collapsed the map run to 2.33: each term opened a cheaper way to earn than the behaviour it was meant to encourage. Potential-based shaping, which cannot change the optimal policy, replaced them. Survival itself is still unsolved: lifespans stay near the random band (251-304 steps).

6. What would settle the rest

More seeds of the honest map on the fixed code and of abl_rssm2, and the potential without the map (on plain DreamerV3) to separate the two. The rule mask and the goal manager, which reach 8-11 achievements on top of RSSM-2, are on family/masked, family/v and family/next.